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220744

4th Sem / Civil

Subject : Soil Mechanics and Foundation Engineering

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Neutral stress refers to _____
- a) Submerged weight of soil
 - b) Pore water pressure
 - c) Saturated weight of soil
 - d) Minor principal stress
- Q.2 If the soil is of expansive type, like in black Cotton Soil, then the best foundation type is _____
- a) End bearing pile b) Friction pile
 - c) Batter pile d) Under-reamed pile
- Q.3 In two phase system soil sample is separated by two separate spaces _____
- a) Solid & Water b) Solid & Air
 - c) Water & Air d) Both a & b

- Q.4 Talus is the soil transported by _____
a) Wind b) Water
c) Glacial d) Gravitational force
- Q.5 Which of the following is cohesive soil?
a) Kankar b) Black cotton soil
c) Loose coarse sand d) Sand with clay
- Q.6 Consolidation theory was given by _____
a) Skempton b) Terzaghi
c) Rankine d) Westergaard

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 The property of a soil which permits water to percolate through it is called _____ (Moisture content / Permeability)
- Q.8 The most reliable field test to determine the bearing capacity of a soil is _____ (cone penetration / Plate load test)
- Q.9 Soils that are carried and deposited by rivers are called _____ (Laterite soil / Alluvial soil)
- Q.10 If two individual footings are too close as per design, then they should be converted as _____ (Mat raft footing / Combined footing)
- Q.11 The effective stress on the soil mass _____ the voids ratio. (increases/Decrease)

Q.12 The soil particles finer than 2 micron size are referred as _____ particles. (Clay/Pebbles)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Explain any four applications of Soil studies in Civil Engineering.

Q.14 Describe the Atterberg's limits of soils, along with their practical significance.

Q.15 Write a short note on pits and trenches methods of soil exploration.

Q.16 Describe any four importance of effective stress of soil in engineering problems.

Q.17 Write a short note on "Creep and Heaving of Soils"

Q.18 With the help of phase diagram state following terms:

a) Porosity

b) Water content

Q.19 Enlist the various methods of compaction of soil in the field and describe any one of them.

Q.20 Describe the factors to be taken into consideration for the selection of a type of foundation.

Q.21 Describe the significance of consolidation of soils and write the name of its types.

- Q.22 Determine the void ratio and degree of saturation for a sample of silty clay having a volume of 14.39 cm^3 , a total mass of 29.03 gm, a dry mass of 24.96 gm, and a specific gravity of 2.67.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 a) Describe the significance of shear strength of soils
- b) Write the different classification of piles and describe any two of them with the help of diagram.
- Q.24 a) Describe the use of “Geo-synthetics” for the improvement of bearing capacity of soils.
- b) Describe any five various factors affecting the permeability of soils.
- Q.25 a) A Constant head permeability test was carried out on a cylindrical sample of sand 10cm diameter and 15 cm height. 160 cm^3 of water was collected in 1.75 minutes under a head of 30 cm. Compute the coefficient of permeability and the velocity of flow.
- b) Describe the process of soil exploration using “Auger Boring method”.